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SECTION 02

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CASE STUDY: ADDICTION IN ONLINE SHOPPING AMONG MALAYSIANS

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1.0 Introduction

The prevalence of online shopping addiction has become a pressing concern in contemporary society, manifesting in compulsive and excessive purchasing behavior that significantly impacts individuals' financial stability, mental well-being, and social relationships. Despite the convenience and accessibility offered by online shopping platforms, the addictive nature of these platforms has led to detrimental consequences, including financial debt, psychological distress, and strained interpersonal connections. Therefore, we are conducting this research to promote healthier consumption habits in the digital age by collecting data to identify their addiction to online shopping.

In this project, we will be questioning the community from different ages and status about their addiction to online shopping. We get to know their opinion on how pleasurable they are about online shopping platforms. We will get to find out how frequent their buying frequency using online shopping platforms from different ages and status. We also get to identify how much they are spending on online shopping platforms.

We used Google Form to prepare the questionnaire as it is convenient for us to distribute it to the community. This is said because we can use the link to share it via social media like WhatsApp, Telegram and Instagram as they are just one click away to fill in the questionnaire. As a result, we successfully gathered 100 responses from the community. Our expectation for this project is to identify how addicted the community is to online shopping as it can be a financial burden to certain people if they can't control themselves in buying things they don't need.

2.0 Data Collection

In this project, our methodology involved gathering data by implementing a survey through Google Form. The survey was carefully structured into specific sections to comprehensively collect relevant information from the respondents. These sections included details such as the personal information of the respondents, their online shopping habits, feedback on their experiences with purchasing unnecessary items, and the overall monthly spend on online shopping. Through this thorough data collection approach, we aimed to obtain an overview of the participants' behaviors in relation to online shopping activities.

No	Question	Answer	Level of Measurement
1	Gender	• Male • Female	Nominal
2	Age	• Short Answer (Accept Integer Only)	Ratio
3	What is your status?	• Student • Working • Not working • Retired	Nominal
4	Did you find the online shopping experience pleasurable?	• Not at all pleasurable • Slightly pleasurable • Moderately pleasurable • Extremely pleasurable	Ordinal

5	How often do you use online shopping platforms?	• Never • Occasionally • Sometimes • Often • Always	Ordinal
6	How often do you find yourself buying something useless or the thing you already have just because it is cheaper online?	• Never • Occasionally • Sometimes • Often • Always	Ordinal
7	How much do you usually spend for online shopping in a month (RM)?	• Short Answer (Accept Integer Only)	Ratio
8	How many times do you shop online in a month?	• Short Answer (Accept Integer Only)	Ratio
9	How many items are currently in your shopping cart?	• Short Answer (Accept Integer Only)	Ratio

Table 1: List of questions asked in questionnaire

3.0 Data Analysis

3.1 Respondents' Age

[illegible]

key 1|7 means 17 years old

Figure 1 : Stem and leaf diagram of respondents' age

Min	17.00
First Quarter	20.00
Median	43.50
Mean	37.71
Third Quarter	51.00
Max	65.00

Table 2 : Analysis from Stem and leaf diagram of respondents' age

As seen in Figure 1, the age distribution of respondents that have answered our questionnaire are around the range of age 17 to 65. Most of the respondents are of age 20, with

the frequency of 32. There are non respondents aged 25 until 29. Also, for respondents in their 30s, there are only 2 respondents of age 34. The second highest frequency of age range is 50s. Table 1 shows that the mean for respondents' age is 37.71 while the median is 43.50.

3.2 Respondents' Status

Status	Frequency	Relative Frequency
Not working	6	0.06
Student	39	0.39
Working	50	0.50
Retired	5	0.05
Total	100	1.0

Table 3: List of respondents' status and its frequency

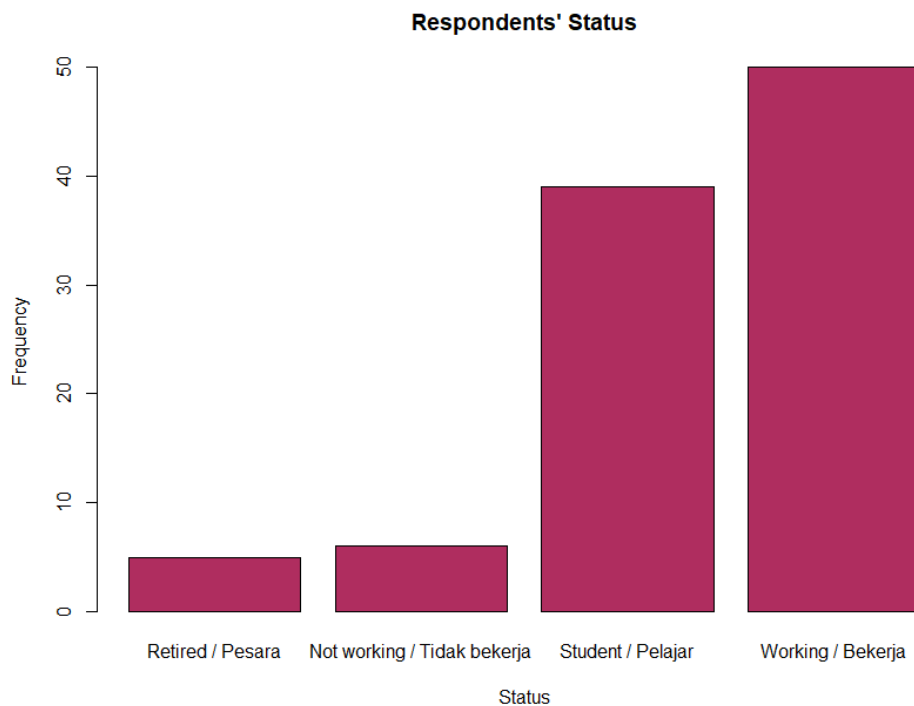


Figure 2: Bar chart of respondents' status

As included in Table 3, we offer four choices of answers to the respondents which are not working, student, working and retired. From the table, we can see the frequency of each status and its relative frequency. From the bar chart in Figure 2, we found that retired people are the least among our respondents which only insist on five people while the working status is the highest. What we can include from this bar chart is that the mode for status is working with a relative frequency of 0.5. This means half of our respondents are from the working category. There are also only 6 respondents that are not working while the rest consist of students with a relative frequency of 0.39.

3.3 Respondents' gender

Gender	Frequency	Percentage(%)
Male	44	44
Female	56	56
Total	100	100

Table 4: Frequency and percentage table of respondents' gender

Respondents' Gender

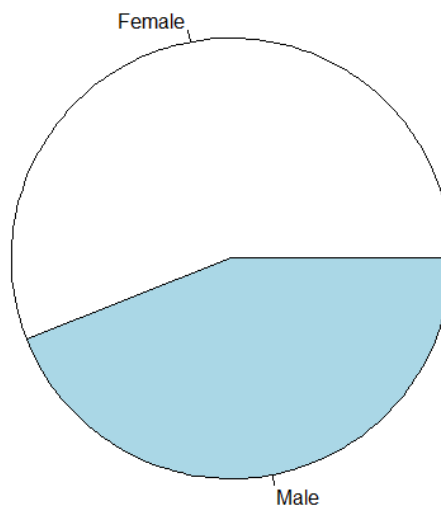


Figure 3: Pie chart of respondents' gender

From the percentage calculated in Table 4, we discovered the majority of our respondents is female with a percentage of 56% although there is only a small difference to the male respondents which include 44% of the total 100 respondents. The white section in the pie chart in Figure 2 indicates female and the blue one presents the male section. The relative frequency of male is 0.44 and female is 0.56.

3.4 Usage of online shopping among the respondents.

Usage of online shopping	Frequency
Always	6
Never	1
Occasionally	21
Often	26
Sometimes	46
Total	100

Table 5: Usage of online shopping frequency table

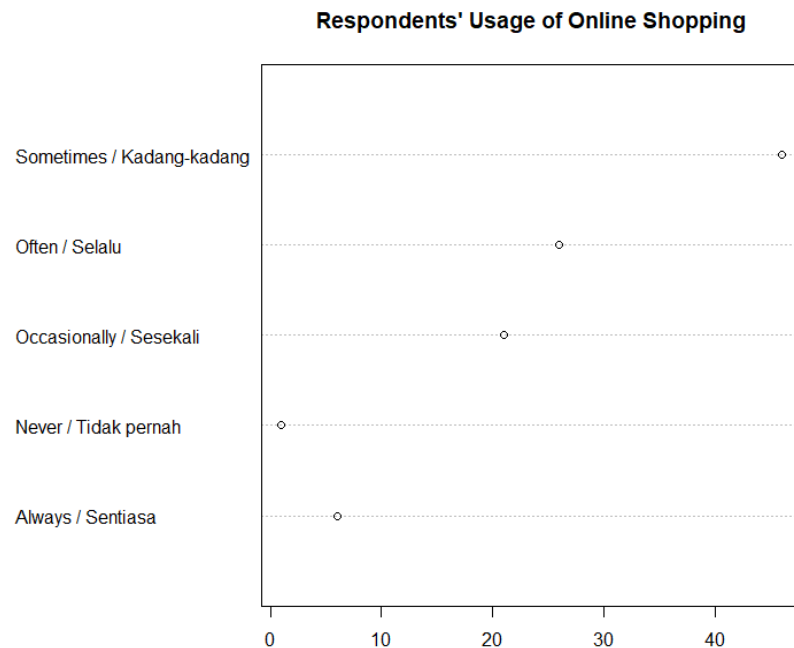


Figure 4: Dotplot of Respondents' Usage of Online Shopping

Based on Table 5, the mode for the usability of online shopping is Sometimes, meaning that most respondents are familiar with online shopping methods but do not use them frequently and only when needed. The lowest goes to Rarely, which means that the percentage of people that are not familiar with online shopping is very little with a percentage of 0.01%. According to R Studio calculations, the median is Occasionally. Only 6 respondents show a trait of an online shopping addict since they admit to always using online shopping platforms. However, 26% of respondents have the early symptoms of being an addict because they often use online shopping platforms.

3.5 Respondents' experience when using online shopping platform

Pleasurable	Frequency	Relative Frequency	Cumulative Frequency (CF)	Percent (%) of CF
Not pleasurable at all	1	0.01	0.01	1.00
Slightly pleasurable	18	0.18	0.19	19.00
Moderately pleasurable	56	0.56	0.75	75.00
Extremely pleasurable	25	0.25	1.00	100.00
Total	100	1.00	100.00	

Table 6: Frequency distribution table of respondents' pleasurable using online shopping platform

Based on the Table 6 provided, it shows how pleasurable the communities' experience in using online shopping platforms is. As shown in the table provided, the majority of respondents reported finding the online shopping experience pleasurable, with a significant proportion finding it either moderately (56%) or extremely (25%) pleasurable. With a small minority (1%) shows that they found it not pleasurable at all. This indicates that the majority of the respondents finds online shopping platforms to be positive and enjoyable due to its convenience in making any purchases and a significant portion finding it to be extremely pleasurable.

3.6 Respondents' buying useless stuff online

Buying Useless Stuff Online	Frequency
Never / Tidak pernah	30
Occasionally / Sese kali	30
Sometimes / Kadang-kadang	26
Often / Selalu	11
Always / Sentiasa	3
Total	100

Table 7: Frequency table of respondents' buying useless stuff online

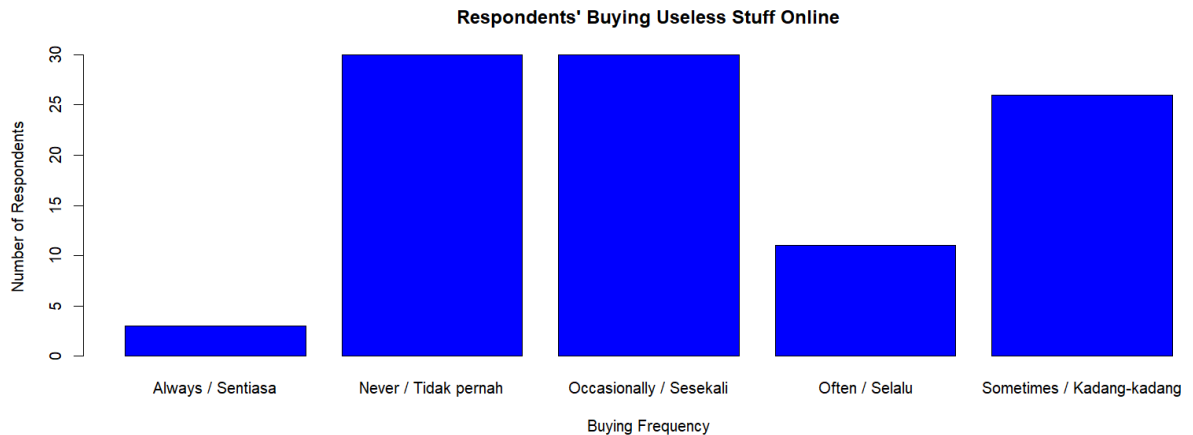


Figure 5: Bar chart of respondents' buying useless stuff online

Based on the Table 7 and Figure 5 shows whether the community is buying useless stuff in online shopping platforms. As shown in the table and figure provided, we offer 5 choices for the respondents, it shows that there are only 3 respondents among 100 respondents that always buy useless stuff online. While there are 11 respondents that often buy, 26 respondents only it sometimes, and lastly there are 30 respondents that buy useless stuff occasionally and 30 respondents never buy useless stuff online. This

indicates that the majority of respondents (60%) either never or only occasionally buy useless stuff online and there are still a significant portion (40%) who do so sometimes, often or always. This also shows that a significant portion lack self-restraint when faced with discounted prices online which could indicate an impulsive buying behavior due to lack of self-control.

3.7 Average Monthly Spending On Online Shopping



Figure 6: Boxplot of the Total of Money That Respondent Spend (RM)

Quartile	Frequency
0.25	50
0.50	100
0.75	150

Table 8: Quartile and frequency of respondents' total spend on online shopping in a month

In Figure 6, the boxplot shows clear distributions of respondents' monthly spending for online shopping and as we can see the box is slightly skewed to the bottom.

From Table 8 our first quartile lies on 50, revealing 25% of the respondents either infrequently online shoppers or choose less expensive products. The median or second quartile of the spending is RM100, which indicates that half of the respondents spend less than RM100. The next quartile, which is the third quartile, is RM150. This means that 75% of the respondents spend RM150 or less per month, showing that most respondents' monthly spending is within an acceptable range. There is also an outlier which is the amount of RM2300 in the data collected which is far above the third quartile.

3.8 Monthly Time Spent On Online Shopping

Monthly Time Spent On Online Shopping	Frequency
0	1
1	34
2	27
3	11
4	9
5	5
6	1
7	2
8	1
9	0
10	7
11	0
12	0
13	0
14	0
15	2
Total	100

Table 9: List Of Monthly Online Shopping Frequency Data

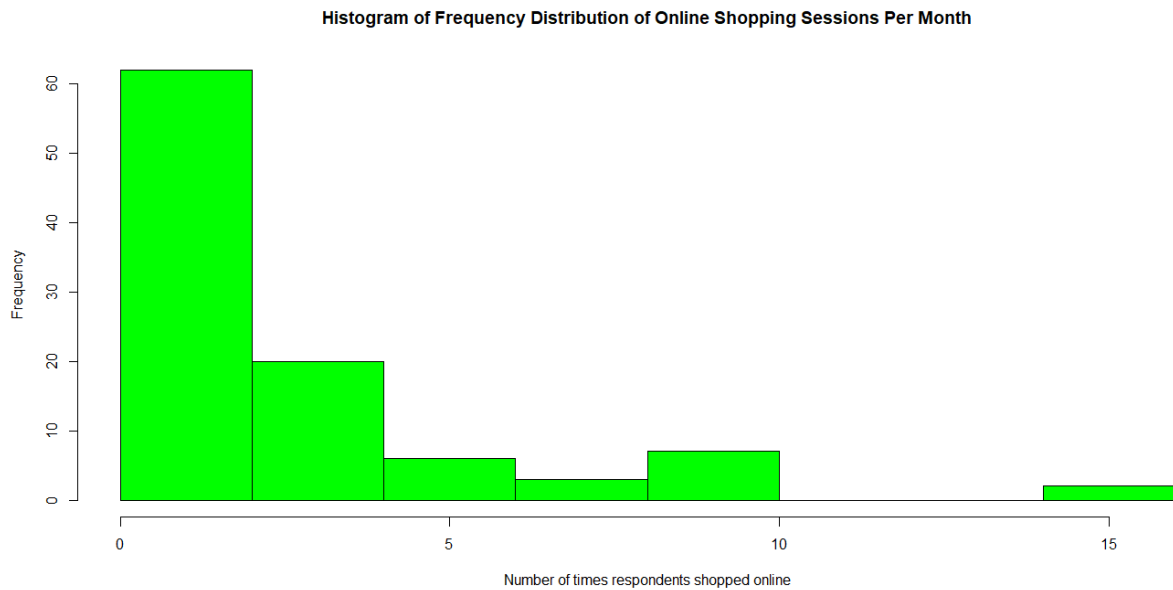


Figure 7: Histogram of Respondents Monthly Time Spend on Online Shopping

Table 9 and Figure 7 show the distribution of respondents' online shopping frequencies, which range from 0 to 15 per month. The histogram in Figure 8 shows that a majority (62%) of respondents purchase online 0 to 2 times per month, indicating infrequent use of online shopping platforms. More specifically, 34% of respondents shop online only once a month, while another 26% shop twice a month, indicating a customer preference for infrequent online purchases. Moreover, 20% of the respondents shop online 3 or 4 times a month indicating a more regular but still moderate use of online shopping platforms. A smaller percentage of respondents which is 6% of them said they went shopping five to six times a month and there is 3% of them shop online up to 8 times in a month. Furthermore, 7 responders, or 7% of the sample, said they shop online 8 to 10 times every month, indicating that there is a certain percentage of people who shop online more frequently. Two participants, or 2% of the respondents, said that they made up to fifteen online purchases per month, which is at the higher end of the frequency range.

3.9 Items in shopping cart

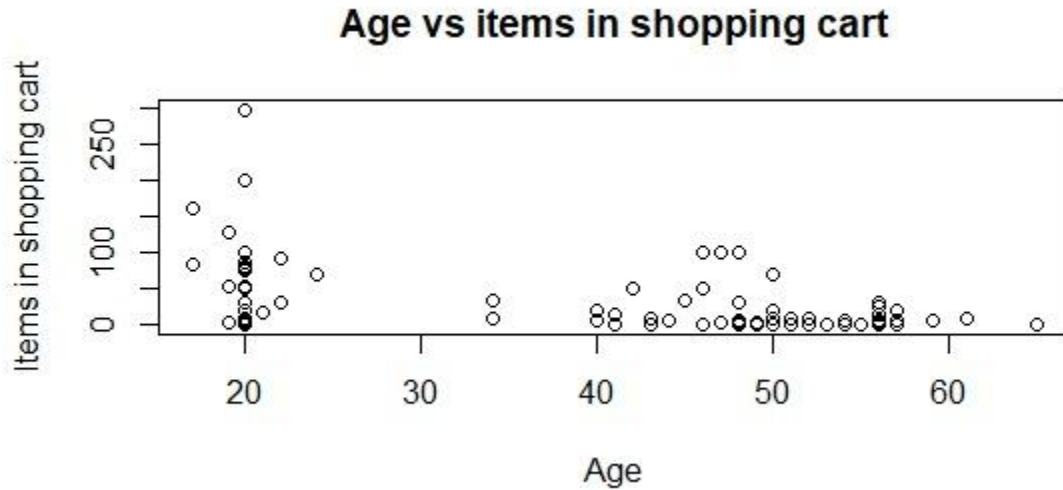


Figure 8: Scatter Plot for Age and Items in Shopping Cart

Items in shopping cart	
Min.	: 0
1st Qu.:	2
Median	: 8.50
Mean	: 30.28
3rd Qu.:	34.25
Max.	: 298

Table 10 : Summary Data of Items in Shopping Cart

Figure 8 shows the relation of respondents' age and the items they put in their shopping cart. From the Figure 8 above, we can see that younger people who are in their 20s put a lot of items in their shopping cart compared to older people. As we can see in Figure 8, most teenagers of age below 25, have 50 to 100 items in their shopping cart. Meanwhile, the respondents of age 40 and above, mostly have 0 items in their shopping cart and also below 50 items.

There are many reasons why younger people have a lot of items in their shopping cart compared to older people. First of all, younger people may live in bigger households or have more people to buy for when they go shopping, including roommates or family members. Second, a wider variety of goods can be found in the carts of younger generations since they are frequently more willing to experiment with new trends and products. A fuller shopping cart might also result from younger people's potential different interests and purchasing habits, such as investing in fashion, technology, or personal hygiene items.

Other than that, elderly people might favor conventional shops and smaller purchases, whereas younger people might be more inclined to embrace internet shopping and bulk purchasing alternatives, which could lead to larger shopping carts.

4.0 Conclusion

Our research has concluded with 100 respondents from different age group ranging from 17 to 65, consisting of 44 males and 56 females. The mode age among our respondents lies at the age of 20 with frequency of 32. The survey we prepared asks the respondents questions revolving around the use of online shopping. Questions such as how often do you use online shopping platform, did you find the online shopping experience pleasurable, online shopping expense in a month, etc are among the important questions that allow us to make deductions and generalize the addiction to online shopping.

To summarize the data we collected, we highlight the significant findings gathered from our survey. During the data collection phase, we learned how to use Google Forms to easily ask questions to our targeted audience. To increase the number of responses, we shared the survey link on social media platforms such as WhatsApp and Telegram. Next, we learned how to make charts with R programming. Using R programming, we can easily create graphs like bar charts, box plots, and histograms with simple coding. Additionally, it can perform instant calculations such as mean and quartiles. The charts assist our data processing process by providing clear visual representations of the obtained data.

We hope that by conducting this research will benefit us in the future by improving our communication and soft skills, making us a more competitive employee if we get assigned with gathering data for a company. Additionally, we believe that by actively participating in the full data collecting and analysis process, we will be able to communicate easily with our colleagues about processes that we need to handle as we have experienced it. Lastly, we would like to express our gratitude to Dr. Noorfa Haszlinna binti Mustaffa, our lecturer, for guiding us on this project and to the all of our respondents who helped in the completion of the research.